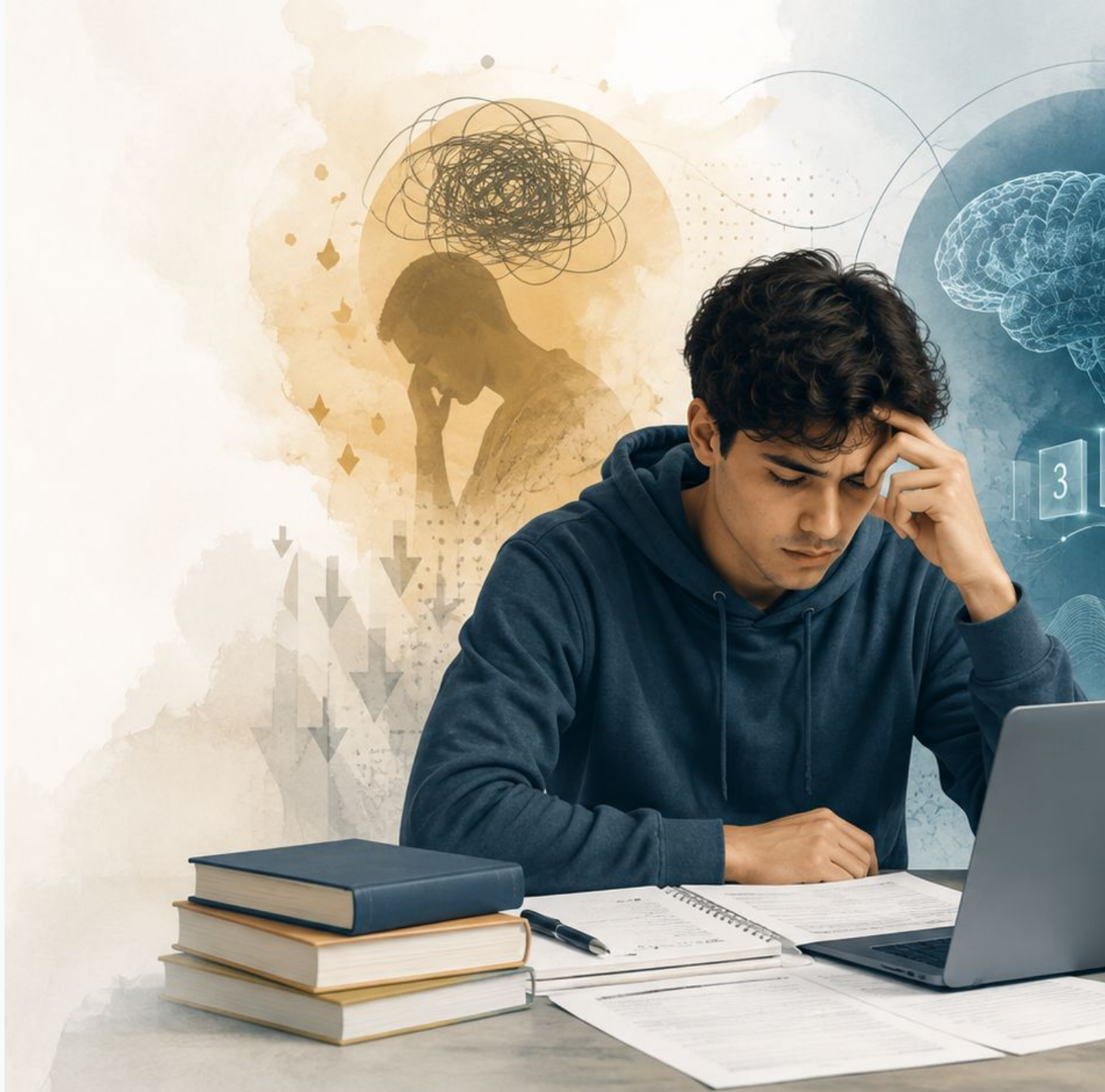
The Relationship Between Academic Stress and Short-Term Memory in Undergraduate Students

PSY101 Research Project

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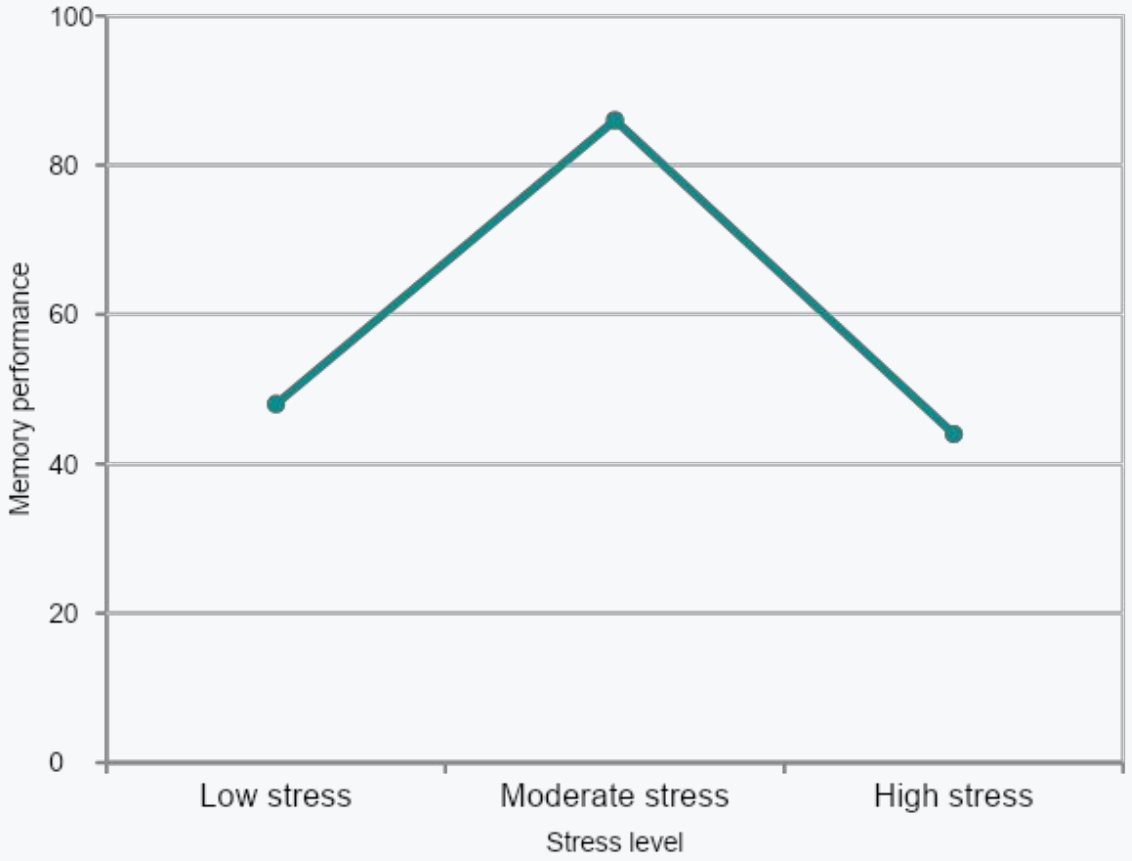
Stress × Memory

Yerkes–Dodson



Why this study?

Stress is common in university life, while digit span provides a simple measure of immediate verbal memory.



Theory

Yerkes–Dodson: performance may be strongest at moderate activation and weaker at very low or very high levels.

Memory model

Forward digit span mainly reflects immediate verbal storage plus attention rather than manipulation of information.

Why a curved test matters

A simple correlation can miss an inverted-U, so the analysis also uses a quadratic regression.

Research question and design

Research question

What is the relationship between academic stress and short-term memory performance in undergraduate students, and is it consistent with an inverted-U pattern?

1 Measure stress

Academic stress over the previous 2 weeks + current stress immediately before the memory task.

2 Measure memory

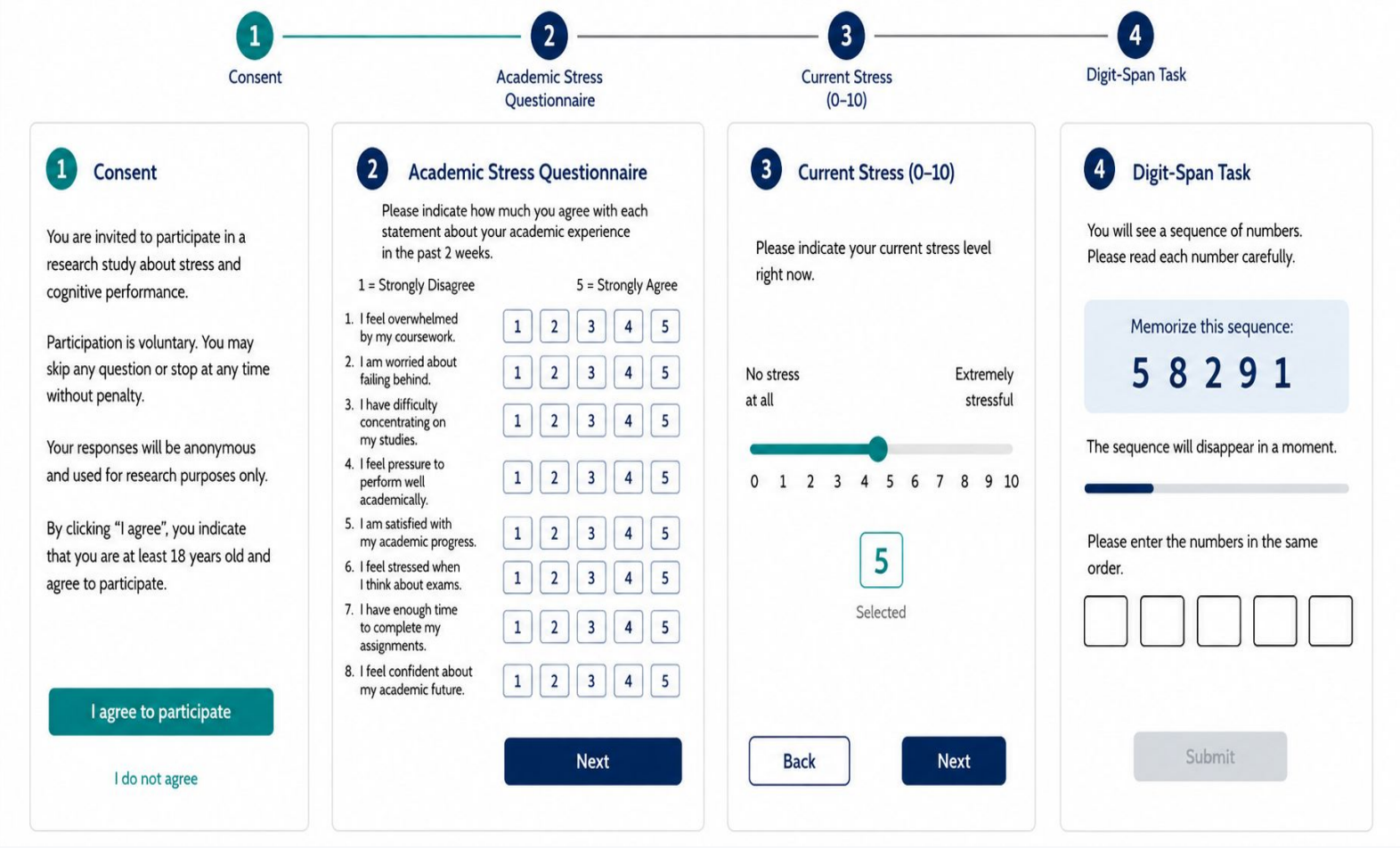
Forward digit-span performance using highest span passed and total correct trials.

3 Test the shape

Pearson correlation for linear association + quadratic regression for the inverted-U prediction.

Design: quantitative • cross-sectional • correlational • no stress induction

Data collection



Participant flow

1. Read consent + agree voluntarily
2. Answer 8 academic-stress items (1–5)
3. Rate current stress (0–10)
4. Complete timed forward digit-span trials
5. Submit anonymous record to a database/CSV

Ethics: no names, student IDs, phone numbers, or forced participation.

What a participant did?

A. Stress measures

Academic stress

Eight researcher-developed statements about exam pressure, workload, time pressure, fear of poor performance, difficulty managing coursework, and feeling overwhelmed. Each item: 1 to 5.

Current stress

One immediate rating from 0 to 10 just before the memory task.

Academic-stress score = sum of 8 items (range 8–40). Higher score = more stress.

B. Forward digit span

Example sequence

5 8 2 9 1

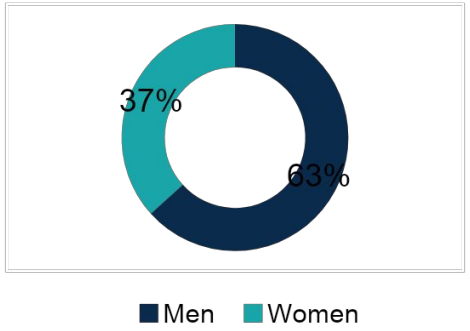
- Digits are presented at a fixed pace (about 1 digit/second).
- Participant recalls them immediately in the same order.
- Sequence length increases across trials.
- No writing digits down or using another device.

Outcomes: highest span passed + total correct trials

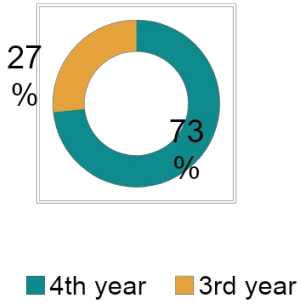
Data analysis

There were 60 participants

Gender



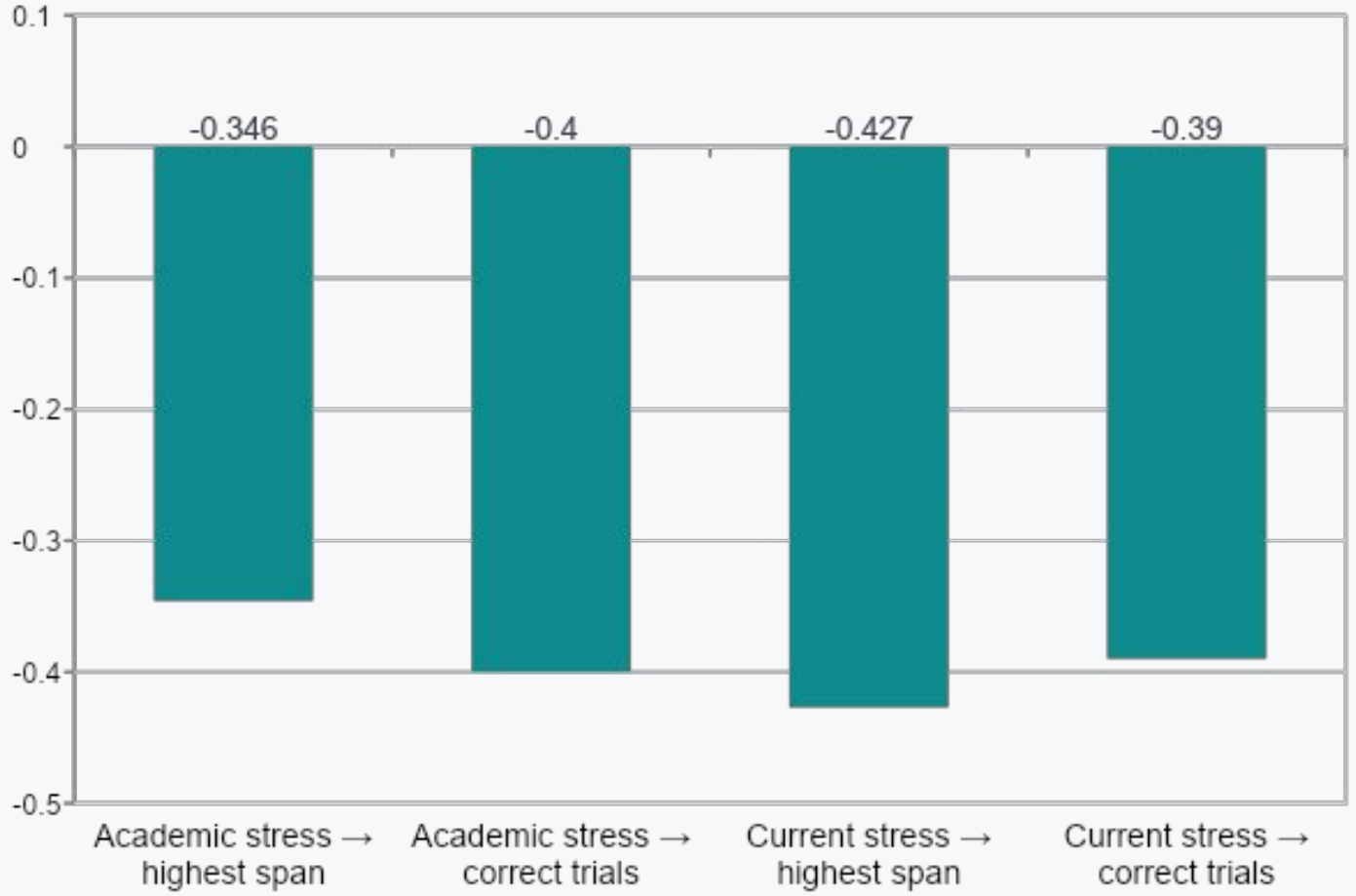
Study year



Age: M = 23.95, SD = 0.70

Largest department group: CSE (60%)

Linear relationships in the analysis dataset



Key values

Academic stress → highest span
r(58) = −.346, p = .007

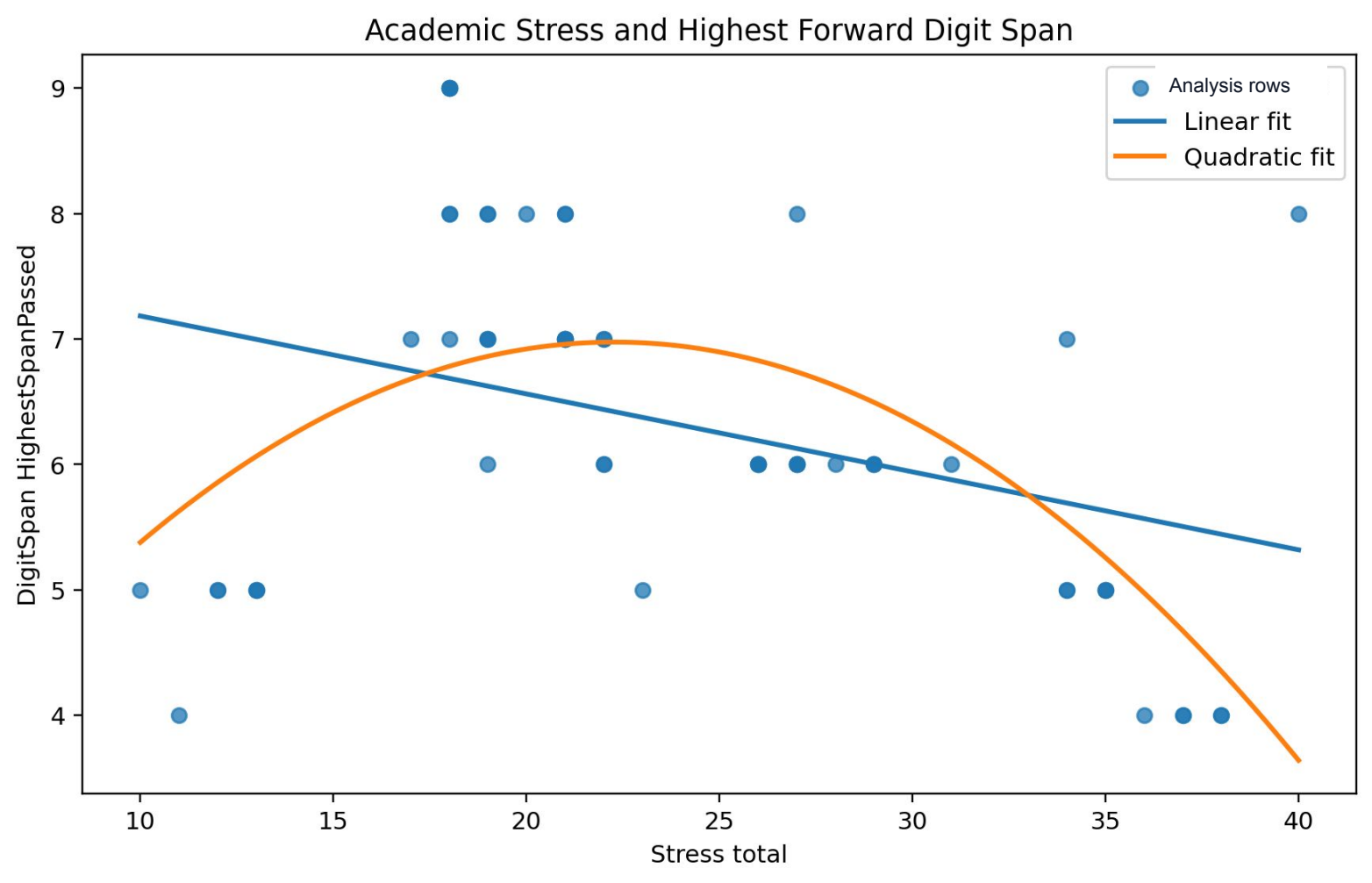
Academic stress → correct trials
r(58) = −.400, p = .002

Current stress → highest span
r(58) = −.427, p < .001

All four correlations were negative and statistically significant in this dataset.

Main analysis: inverted-U pattern

Academic stress predicting highest forward digit span

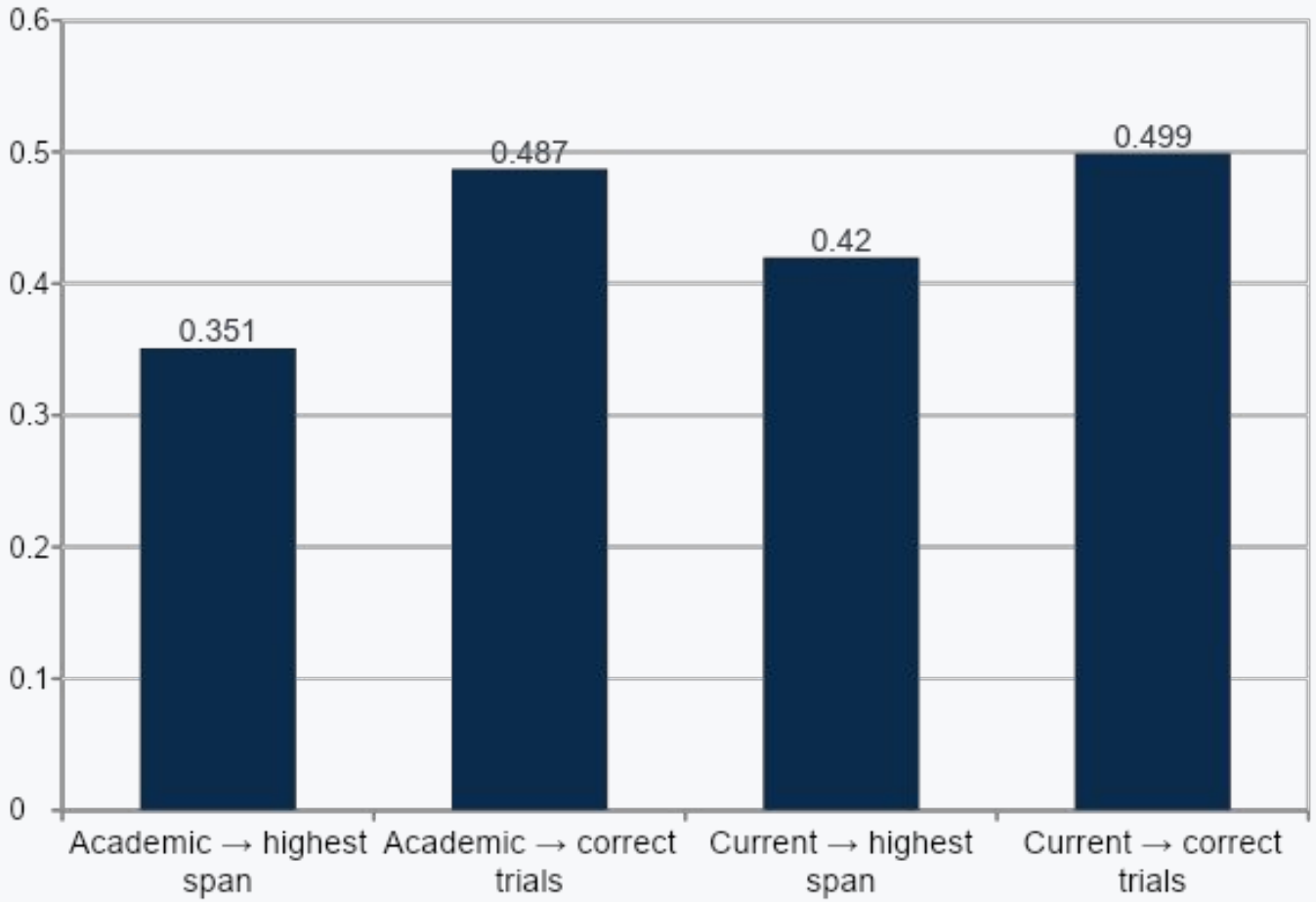


Quadratic model

R^2	**.351**
Adjusted R^2	**.328**
Squared term b	**−0.0106**
p for b^2	**< .001**
ΔR^2 vs linear	**.231**
Estimated peak	**22.27**

Interpretation: negative b^2 = downward curvature; the fitted peak is around moderate stress in this dataset.

The curved pattern appeared across all four models



Squared terms

Academic / highest	b = −.0106	p < .001
Academic / correct	b = −.0222	p < .001
Current / highest	b = −.1217	p < .001
Current / correct	b = −.2584	p < .001

What the results mean — and what they do not mean

Interpretation

• Linear analyses: more stress is associated with poorer digit-span performance.

• Quadratic analyses: the fitted relationship is strongest around moderate stress and weaker toward the extremes.

• This is statistically consistent with an inverted-U interpretation.

Limits / caution

• Correlation ≠ causation.

• Researcher-developed stress items are not a validated clinical scale.

• Peak values are sample-specific, not “ideal stress targets.”

• Convenience sampling and a narrow age/year profile limit generalization.

Conclusion

Higher stress was associated with poorer digit-span performance in the linear analyses, while quadratic models described the relationship better and showed a downward-curving pattern consistent with the Yerkes–Dodson framework.

Takeaway

The shape of the relationship matters—not only the correlation.

Practical idea

Reduce excessive academic pressure and support sustainable study routines.

Future study

Collect genuine responses from a larger, more diverse sample using a standardized digital workflow.

Thank you